

Vitor Gama

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EDUCATION

West Virginia University

Ph.D. student, Chemical Engineering;

GPA: 3.75/4.00

Morgantown, WV, USA

Aug. 2021 – Dec. 2025 (Expected)

Federal University of Campina Grande

M.Sc., Chemical Engineering;

GPA: 9.14/10.00

Campina Grande, Paraiba, Brazil

Sept. 2018 – April 2021

Federal University of Campina Grande

B.Sc., Chemical Engineering;

GPA: 8.18/10.00

Campina Grande, Paraiba, Brazil

May 2013 – Aug. 2018

AWARDS & ACHIEVEMENTS

- **AIChE Environmental Division Graduate Paper Award (2024):** Awarded 2nd place with the paper titled “Process Operability Analysis of Membrane-Based Direct Air Capture for Low-Purity CO₂ Production”.
- **Jack and Marietta Mullenger Fellowship (2024)**
- **Statler College PhD Recruitment Fellowship (2021)**

RESEARCH PUBLICATIONS

Complete list available on my [Google Scholar](#).

- [1] **Vitor Gama et al.** “Connecting Material Characteristics with System Properties for Membrane-based Direct Air Capture (m-DAC) using Process Operability and Inverse Design Approaches”, **Industrial Engineering and Chemistry Research**. (under review).
- [2] **Vitor Gama et al.** “Process Operability Analysis of Membrane-Based Direct Air Capture for Low-Purity CO₂ Production”, **ACS Engineering Au**, 4.4 (Aug. 2024), pp. 394-404. DOI: 10.1021/acsengineeringau.3c00069.
- [3] **Vitor V. Gama et al.** “Modeling and Process Operability Analysis of a Direct Air Capture System”, **IFAC-PapersOnLine**, 13th IFAC Symposium on Dynamics and Control of Process Systems, including Biosystems DYCOPS 2022, Vol. 55.7 (Jan. 2022), pp. 316-321. DOI: 10.1016/j.ifacol.2022.07.463. ISSN: 2405-8963.

CONFERENCE PRESENTATIONS

- [1] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Leveraging Process Operability Mapping to Support Experimental Membrane Direct Air Capture (m-DAC) Solutions*. 2024 AIChE Annual Meeting, San Diego, CA, October 29, 2024 (Oral Presentation)
- [2] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Process Operability Analysis of Membrane-Based Direct Air Capture for Low-Purity CO₂ Production*. Membranes: Materials and Processes – Gordon Research Conference, New London, NH, July 28 – August 2, 2024 (Poster Presentation)
- [3] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Evaluation of Hollow Fiber Membrane Modules for CO₂ Direct Air Capture and Utilization in Low-Purity CO₂ Processes*. 2023 AIChE Annual Meeting, Orlando, FL, November 7, 2023 (Oral Presentation)
- [4] **Vitor Gama**, San Dinh, Oishi Sanyal, Fernando V. Lima. *Feasibility Analysis of a Membrane System for Direct Air Capture of CO₂*. 2022 AIChE Annual Meeting, Phoenix, AZ, November 17, 2022 (Oral Presentation)

- [5] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Designing a Membrane-based Platform for CO₂ Direct Air Capture Systems Analysis*. Gordon Research Conference on Chemical Separations, Ventura, CA, October 2–7, 2022 (Poster Presentation)

RESEARCH EXPERIENCE

West Virginia University

Morgantown, WV, USA

Graduate Research Assistant (Ph.D.) [Advisors – Dr. Fernando V. Lima/Dr. Oishi Sanyal]

Aug. 2021 – Currently

- Process design & simulation
- Operability analysis studies to expedite process evaluation and deployment
- Application of operability and simulation frameworks to a gas separation membranes for Direct Air Capture (DAC) with Carbon Capture Utilization (CCU).

Federal University of Campina Grande

Campina Grande, Paraiba, Brazil

Graduate Research Assistant (M.Sc.) [Advisor - Dr. Antonio Tavernard]

Sept. 2018 – April 2021

- Chemical industry equipment cost estimation research based on literature models.
- Developed a Windows OS application for chemical industry equipment cost estimation based on literature models. (“CostApp: a Cost Estimation Tool Developed Using C# and WPF for the Chemical Engineering Field”)

Federal University of Campina Grande

Campina Grande, Paraiba, Brazil

Developer

Sept. 2019 – April 2021

- Technological Project for the Development of a Method for Synthesis of Control structures (PETROBRAS/CENPES/PAQTCPB/LENP/UFCG): An automated software capable of easily selecting the most promising self-optimizing control structures in industrial processes.
- Worked on coding the PID routines to control simulated processes uploaded to the BRPWC tool
- Worked on developing mock-ups for the process control user interface of the tool and code troubleshooting

SKILLS

Programming: Python, MATLAB, Markdown, C#, Javascript, LaTeX and exposure to Java

Technologies/Platforms: Git, GitHub, GitLab

Process simulation: Aspen Plus, Aspen Custom Modeler, HYSYS, AVEVA Process Simulation, PRO/II, ChemCad

Languages: English and Portuguese

RELEVANT COURSEWORK

Major coursework: Transport Phenomena, Advanced Chemical Engineering Thermodynamics, Chemical Reaction Engineering, Mathematical Methods in Chemical Engineering, Teaching Practicum

Minor coursework: Dynamic Simulations, Membrane Separations, Artificial Intelligence Techniques

PROFESSIONAL LEADERSHIP AND SERVICE

Brazilian Student Association (President)

Sept. 2022 – Sept. 2024

Statler College DEI Student Committee

May 2022 - May 2023

CODES Research Group Leader

Sept. 2022 - Present

CLIFTONSTRENGTHS

Arranger | Positivity | Input | Woo | Communication